

DraftRun Pipeline AS IS

Current as of Slice 2.17.4.6.1.3.9.1: Alternative-Angle Route Dossier Budget Repair.

This document is the maintained technical map of the current DraftRun generation pipeline. It describes the running system as it exists now, not the target design. Update this file and regenerate the PDF whenever a slice changes DraftRun steps, artifacts, role model usage, context flow, retry/fallback behavior, validation, ranking, revision, or trace semantics.

PDF quick-view copy: docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.pdf.

Regenerate PDF:

```
python scripts/generate-draft-run-pipeline-pdf.py
```

How AS IS Participates in DoD

This document is not only a trace map. For any complex DraftRun slice it is an acceptance contract:

1. Before planning or implementation, the slice DoD must cite the AS IS sections it preserves, changes, or intentionally supersedes.
2. Provider-heavy changes must derive their requirements from the current step order, role handoff, context flow, payload budget, operation envelope, runtime budget, and quality/fidelity rules in this document.
3. The slice DoD must name the runtime evidence that will prove the behavior: step artifact path, child AiRun.requestPayload, operationEnvelope, payloadBudget, runtimeBudget, qualityFidelity, diagnostics script output, or known debt entry.
4. At the end of the slice, the implementer must state either "AS IS unchanged" with the reason, or update this Markdown and regenerate docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.pdf.
5. If the current AS IS behavior is known to be weaker than the target design, the document must say so as known debt and link the repair slice or TO BE/ADR. Do not describe a target state as if it already exists.

Minimum DoD input from this document:

- step order and persisted artifact handoff;
- which role/model receives which provider projection;
- whether the current provider call has direct budget proof;
- where provider retry, repair, backup, fallback, not-run, and incidents are visible;
- whether DraftRun.status=succeeded is enough for the claim being tested. It is not enough for editorial quality; use qualityFidelity.

1. Core concepts

Concept	Meaning	Stored where
DraftRun	Parent orchestration run for one approved post fabula. It owns step order, status, final draft, and child AiRun ids.	var/glavred-draft-runs.sqlite3

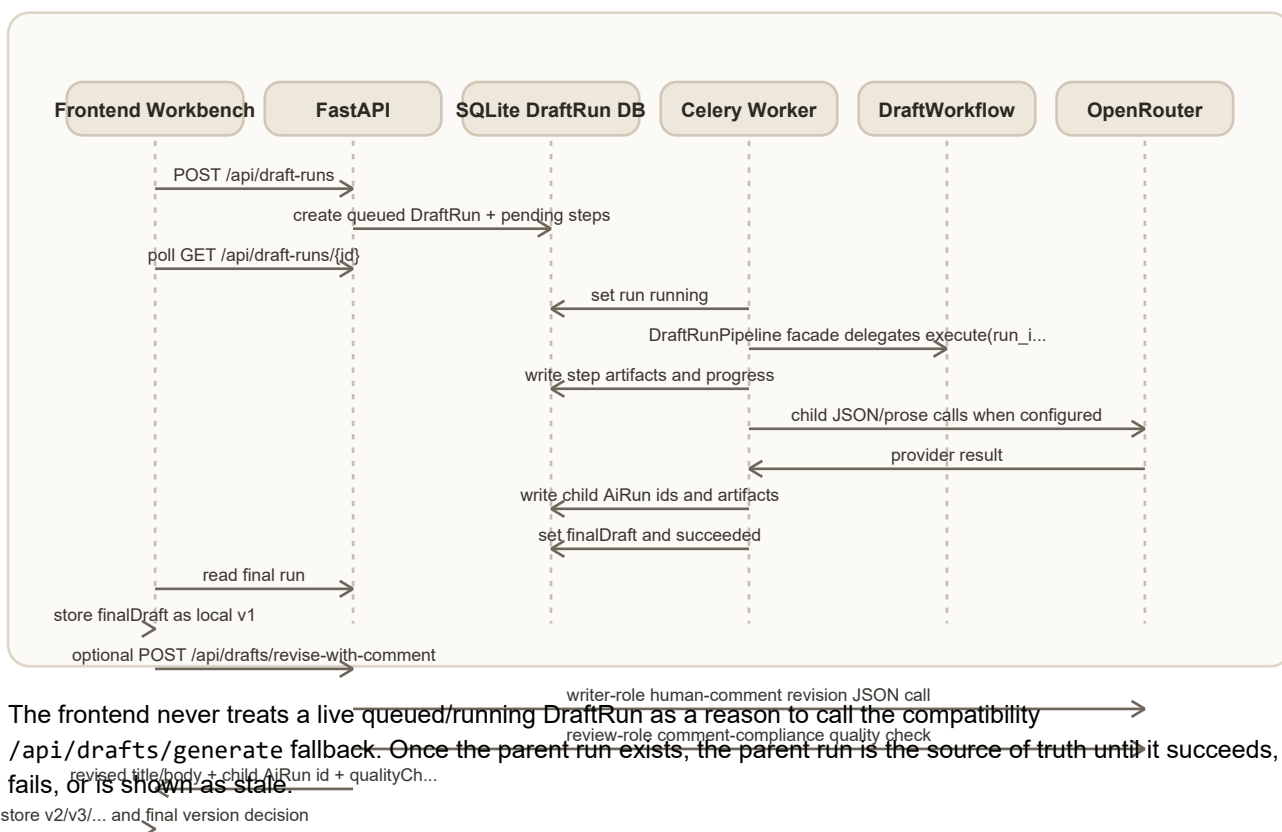
Concept	Meaning	Stored where
DraftRunStep	One logical pipeline stage: context, source intent, public evidence, feasibility, contract, rules, planning, candidates, validation, complete.	draft_run_steps.artifact_payload
AiRun	One model/provider call or audited fallback inside a DraftRun.	var/glavred-ai-runs.sqlite3
SourceLedger	Claim/provenance inventory with allowed use, confidence, warnings, risks, and forbidden inferences.	context and enriched publicEvidence artifacts
EvidenceInterpretation	Editorial interpretation of accepted evidence: implications, tensions, usable examples, limits, forbidden overclaims, reader-value hooks, and rejected uses.	rulePack.evidenceInterpretation, child AiRun trace
PostContract	Locked editorial intent: thesis, audience, CTA, allowed claims, forbidden moves, size contract, topic/fabula obligations.	postContract artifact
RuleRegistrySnapshot	Machine-readable rules and validator bindings derived from contract, ledger, topic, fabula, and publisher rules.	rulePack.ruleRegistrySnapshot
ArticleDossier	DraftRun-local article memory built from artifacts. It is not workspace persistence and not vector storage.	selected step artifacts
ContextPack	Role-specific subset of ArticleDossier passed to LLM calls.	step artifacts and child AiRun.requestPayload
EditorialCritiqueReport	Report-only prosecutor/editor critique of candidate idea strength, tension, author stance, source integration, and reader value.	validation.editorialCritiqueReport, child AiRun trace
AlternativeAngleTournament	Critic-driven challenger route and candidate used to escape a weak local optimum.	validation.alternativeAngleTournament, child AiRun trace
RevisionLoop	Bounded editorial optimization loop: validator repair goals plus editorial goals, old-vs-new dimension scoring, accepted/rejected revisions, and rejected moves.	validation.rankingRevision.revisionLoop
FinalQualityGate	Last machine acceptance layer for the delivered final draft as public prose; combines deterministic checks with an independent final-gate model review and may run bounded writer repair cycles if actionable gate findings require it. It separates real missing attribution from diagnostic attribution handoff noise.	validation.rankingRevision.finalQualityGate
DraftRunBudget	Effective research/execution caps derived from Fabula.researchDepth and backend execution mode.	context.draftRunBudget, downstream budget metadata
PayloadBudget	Provider-input boundary for LLM calls: operation profile, execution mode, prompt/token estimates, sent/trimmed counts, suppressed fields, semantic inputs, quality risk, and over-budget incidents.	child AiRun.requestPayload.payloadBudget, attempts, operationEnvelope.payloadStats
ProviderDossier	Typed, operation-family projection assembled from rich artifacts through deterministic context access. It records required/optional inputs, actual sent data, handles, counts, suppressed fields, readiness, and quality risk. Planning, writer/alternative-angle, review, ranking, revision, and final-quality runtime call sites are connected; Slice 3.9 is live-accepted by run 7bf3a7b9-4646-4b32-8bfd-cc5b200a1b47.	provider-free replay/audit output; migrated child AiRun.requestPayload.providerDossier and providerInput

Concept	Meaning	Stored where
ValidationRuntimeBudget	Runtime boundary for the validation/revision loop: wall-clock, LLM-call, revision-cycle, pairwise-round, final-repair, non-improvement, heartbeat, slow-but-healthy, and canonical stop-reason accounting.	validation.progress.runtimeBudget, validation.rankingRevision.runtimeBudget, validation.rankingRevision.revisionLoop.runtimeBudget
ProviderOperationRuntime	Step-local progress boundary for a currently running provider operation outside the validation runtime loop. It records operation id, operation kind, operation start time, model role, selected model, prompt/token estimate when available, provider wait seconds, stale budget, and slow-but-healthy state.	steps[].artifactPayload.progress, scripts/analyze_draft_run_reliability.py runtime diagnostics
SQLiteRuntimeStorageGuard	Local durability/error-handling guard for DraftRun and child AiRun SQLite stores. It applies timeout, busy-timeout, local WAL or Docker-bind DELETE journal mode, normal synchronous mode, foreign keys, controlled connection open/commit/rollback, integrity checks, and storage diagnostics for malformed/unavailable DB files.	backend.app.infrastructure.sqlite_runtime, SQLITE_JOURNAL_MODE, scripts/check_sqlite_integrity.py
QualityFidelityReport	Final per-run distinction between technical completion, provider retry/backup/fallback recovery, evidence fidelity, validation/final-gate issue lifecycle, editorial publishability, and overall clean/degraded/attention verdict. A succeeded DraftRun is not trusted quality by itself: open critical issues block clean success, and unresolved final-gate warnings can only be publishable with caution.	validation.rankingRevision.qualityFidelity, complete.qualityFidelity, diagnostics helper output
ProviderReliabilityReport	Cross-run provider reliability summary and remediation map for retry, backup, fallback, timeout, malformed JSON, schema failure, payload/runtime budget, degraded, failed, and open critical patterns.	python scripts/analyze_draft_run_reliability.py --run-id ...
finalDraft	The selected draft returned to the frontend after validation, ranking, revision loop, and final quality gate.	parent DraftRun
DraftVersion	Immutable editor-facing version of the delivered draft. v1 is the machine final draft; later versions come from human-comment revisions or manual edits.	local workspace postDraft.versions[]
HumanCommentRevisionQualityCheck	Diagnostic review of one human-comment revision: comment compliance, source-marker preservation, public-prose health, internal jargon leaks, and base-version regression risks. It never blocks saving the version.	local workspace postDraft.versions[].qualityCheck, child AiRun trace
EditorDecisionSnapshot	Human final-selection record linking the chosen version to machine trace summaries, unresolved risks, comments, and manual edit counts.	local workspace finalText.editorDecisionSnapshot
EditorialLearningAuthorNote	Auto-created author-memory note summarizing what the editor appears to have taught the system through final version choice, comments, manual edits, rejected versions, and quality checks. Starts as pendingReview; only accepted notes influence author-position inference.	local workspace authorNotes[], type editorialLearning

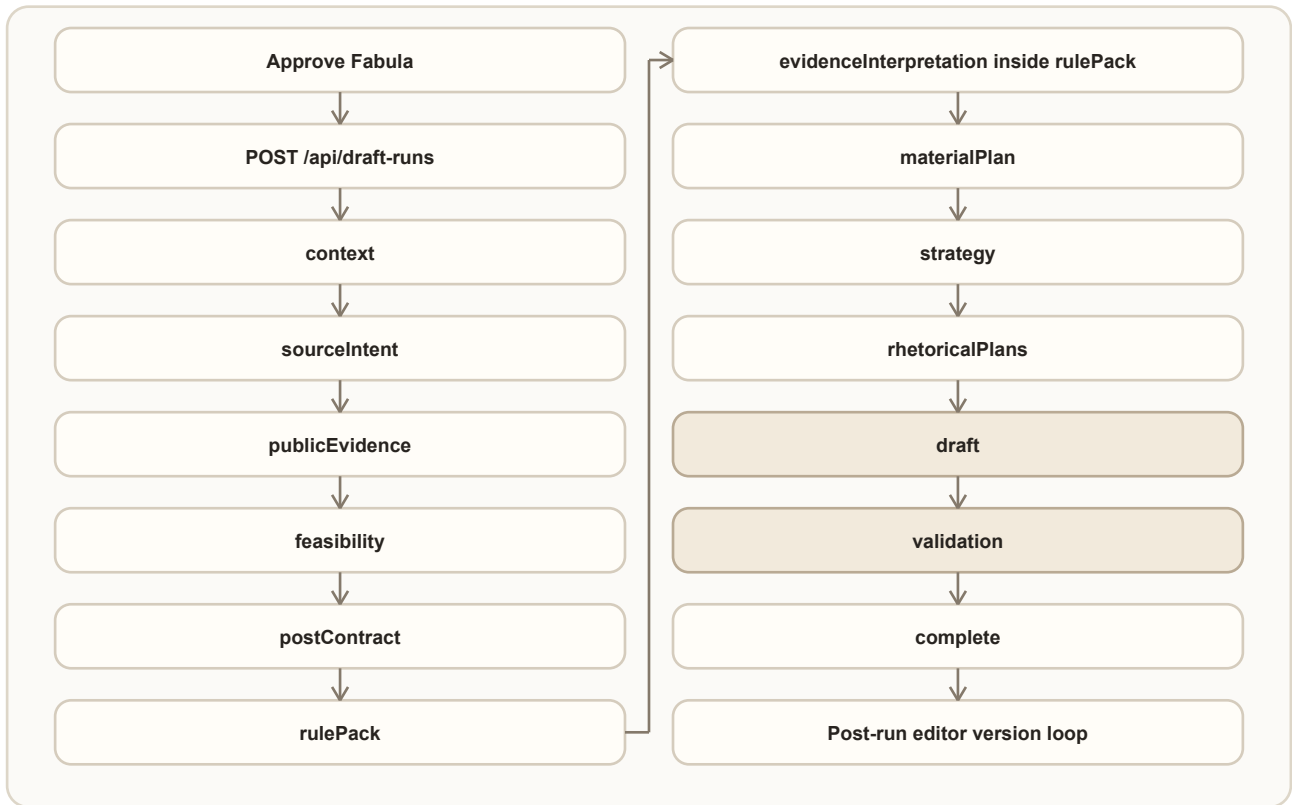
2. Runtime topology

Runtime entrypoints are compatibility-preserving:

- the Celery/API path still enters the drafting runtime through `backend.app.application.draft_run_pipeline.DraftRunPipeline;`
- `DraftRunPipeline` is now a thin facade that keeps the previous constructor and `execute(run_id)` method;
- sequencing is delegated to `backend.app.drafting.application.workflow.DraftWorkflow` through an ordered `DraftStepRegistry;`
- context/artifact, source/evidence, evidence-contract, planning, generation, validation, revision, final-quality, and HITL services now have canonical owners under `backend.app.drafting.application;`
- old `backend.app.application.*` imports for migrated runtime modules are thin compatibility shims only;
- prompt text, provider calls, artifact payloads, progress semantics, child `AiRun` persistence, and blocked/failure behavior are unchanged.



3. Current step order



The backend enum is:

context -> sourceIntent -> publicEvidence -> feasibility -> postContract -> rulePack -> materialPlan -> strategy -> rhetoricalPlans -> draft -> validation -> complete

The ordered runtime registry uses the same sequence. It is a package-boundary refactor, not a semantic step change.

4. Step-by-step technical flow

The role system is not a separate chat room. Roles interact through persisted DraftRun artifacts. Each LLM call chooses a role model through DraftModelRoleResolver, receives the artifact set and the role-specific ContextPack, writes a child AiRun, and stores its result back into the parent step artifact. The next role reads that artifact instead of relying on hidden conversation state.

Role-aware execution map:

Step	Active role	Model selection	Context passed to the role	Output consumed by
context	none	no model	frontend snapshot only	source intent, ledger, feasibility, DraftRun budget
sourceIntent	research	DRAFT_RESEARCH_MODEL, then default	brief sources, context summary, initial ledger, DraftRun budget	public evidence
publicEvidence search	web search	OPENROUTER_WEB_SEARCH_MODEL	budget-capped research tasks and built search query	evidence synthesis
publicEvidence synthesis	research	DRAFT_RESEARCH_MODEL, then default	accepted public evidence, initial ledger, context	enriched ledger, dossier
feasibility	none	no model	enriched ledger and warnings	contract or blocked completion

Step	Active role	Model selection	Context passed to the role	Output consumed by
postContract	none	no model	brief, ledger, feasibility, size settings	rules, planning, validation
rulePack	none	no model	contract, ledger, publisher/topic/fabula rules	evidence interpretation, validation, revision
evidenceInterpretation inside rulePack	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	compact enriched ledger, evidence synthesis, accepted public evidence, contract, relevant rule subset, strategy ContextPack, timeout envelope	material plan, dossier, context packs
materialPlan	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	strategy ContextPack, usable evidence candidates, evidence interpretation, contract, registry	draft strategy
strategy	strategy	DRAFT_STRATEGY_MODEL, then default	material plan, rules, contract, strategy ContextPack	rhetorical plans
rhetoricalPlans	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	strategy, material plan, claim/rule ids	writer candidates
draft	writer	DRAFT_WRITER_MODEL, then default, repair, backup, domain-safe fallback	writer ContextPack, one rhetorical plan, material evidence, contract, writer generation params	validation and ranking
validation lint	none	no model	candidates, contract, registry, ledger	LLM validation and ranking
validation LLM review	review	DRAFT_REVIEW_MODEL, then default, repair, backup	review ContextPack, candidates, deterministic findings	pairwise ranking
validation editorial critique	critic	DRAFT_CRITIC_MODEL, then default, repair, backup	critic ContextPack, candidates, evidence interpretation, validation findings	trace, dossier, future critique-aware ranking
validation alternative angle	anotherAngle then writer	DRAFT_ANOTHER_ANGLE_MODEL for route, DRAFT_WRITER_MODEL for challenger prose, repair, backup, no fake challenger fallback	initial validation, editorial critique, another-angle ContextPack, writer ContextPack, another-angle/writer generation params	final validation and ranking
validation ranking	review	DRAFT_REVIEW_MODEL, then default, repair, backup	merged candidates, old scorecard, deterministic and LLM findings	revision loop
validation revision	writer	DRAFT_WRITER_MODEL, then default, repair, backup, failed/not-run if no usable JSON	current best candidate, repair goals, anti-regression constraints, writer ContextPack, revision generation params	regression guard and final draft
validation final quality gate	finalGate, then optional writer	DRAFT_FINAL_GATE_MODEL, or independent critic/review/default fallback; DRAFT_WRITER_MODEL for bounded final repair cycles	delivered final candidate, FinalQualityContract, validation reports, critique, evidence interpretation, ledger, contract, rule registry, material plan	accepted public draft or rejected repair trace
complete	none	no model	final DraftRun state	frontend

Step	Active role	Model selection	Context passed to the role	Output consumed by
post-run human comment revision	writer	DRAFT_WRITER_MODEL, then default, repair, backup, failed if no usable JSON	selected DraftVersion, editor comment, compact machine trace context from final quality gate, revision loop, alternative angle, validation, contract/material evidence	new local DraftVersion; not a new DraftRun
post-run human revision quality check	review	DRAFT_REVIEW_MODEL, then default, repair, backup, notRun if no usable JSON	base version, revised version, editor comment, compact machine trace context, source markers and public-prose constraints	DraftVersion.qualityCheck; child AiRun, not a DraftRun step

Practical reading rule: when a step says Role/model handoff: strategy, it does not mean that the strategy model talks directly to the writer model. It means the strategy role writes a structured artifact, then the writer role later receives that artifact plus its own compact context pack.

4.1 context

Purpose: build the normalized local context for the selected work item.

Input:

- approved PostBrief;
- EditorialModel;
- draftContext snapshot from frontend: work item, plan slot, candidate, signal, topic, fabula, publisher rules, author-position evidence, publication-size context.

Processing:

- build_draft_run_context_summary(...) normalizes the frontend snapshot;
- SourceLedgerBuilder builds the initial internal claim ledger.
- DraftRunBudgetResolver combines fabula.researchDepth with DRAFT_RUN_EXECUTION_MODE.

Output:

- contextSummary;
- initial sourceLedger;
- draftRunBudget;
- missingContext and compatibility metadata when links are absent.

Role/model handoff:

- no LLM role is used;
- this step creates the first artifact boundary for later roles;
- downstream research, strategy, writer, and review calls never read the raw frontend workspace directly after this point.

Trace: steps[].stepKey = context.

4.2 sourceIntent

Purpose: convert approved brief sources into an explicit research plan.

Input:

- `PostBrief.sources`;
- context summary;
- initial `SourceLedger`.

Processing:

- `SourceIntentNormalizer` classifies URLs, named sources, human-language requests, proof needs, framing hints, exclusions, and unknown lines;
- `ResearchPlanService` may call `OpenRouter` through the research role;
- deterministic fallback preserves original wording when provider planning fails;
- `source_research_budgeting` caps executable verification tasks and records `budgetTrace.budgetSkipped` instead of silently dropping them.

Output:

- normalized `sourceIntent`;
- `researchPlan`;
- `draftRunBudget` and `budgetTrace`;
- child `AiRun` when `OpenRouter` is used.

Role/model handoff:

- active role: research;
- model resolver chooses `DRAFT_RESEARCH_MODEL` or falls back to `OPENROUTER_DEFAULT_MODEL`;
- the role receives sources, context summary, and the initial ledger;
- it returns a `ResearchPlan`, not proof;
- `publicEvidence` consumes the plan and decides what can actually be read or searched.

Trace: `sourceIntent` step and child `AiRun.requestPayload.draftRunStep` = `sourceIntentResearchPlan`.

4.3 publicEvidence

Purpose: execute available public evidence tasks and enrich the ledger.

Input:

- `sourceIntent` artifact;
- `researchPlan`;
- current context artifact.

Processing:

- exact URL tasks are read by the URL reader;
- public search tasks call OpenRouter web search only when web tools are enabled;
- retrieval tasks, URL reads, and search result counts are capped by DraftRunBudget;
- disabled or unavailable search tasks remain explicit attempts, not proof;
- relevance guard rejects search-result drift;
- EvidenceSynthesis reconciles accepted public evidence into external claims;
- SourceLedgerExternalEvidenceMerger merges accepted external claims into the ledger;
- accepted evidence items and external ledger claims are trimmed by budget before downstream planning;
- operation progress is persisted during URL/search/synthesis work.

Output:

- PublicEvidenceBatch: attempts, accepted evidence items, warnings;
- EvidenceSynthesis: external claims, decisions, warnings;
- enriched SourceLedger;
- budget metadata: used counts, cap hits, skipped tasks, trimmed evidence/claims;
- initial ArticleDossier and all role ContextPacks.

Role/model handoff:

- public search uses the web-search model/config;
- evidence synthesis uses research.
- web search receives concrete URL/search tasks and built search queries;
- research synthesis receives accepted evidence candidates and decides how they should become external ledger claims;
- downstream roles consume the enriched ledger and dossier, not raw search output.

Trace:

- publicEvidence artifact;
- child AiRun.requestPayload.draftRunStep = externalEvidenceSynthesis;
- child web-search AiRun records when search is enabled.

AS IS limitation:

- public evidence retrieval and synthesis do not write prose directly. Accepted evidence is merged into the ledger here, then interpreted inside rulePack before material planning and writing.

4.4 feasibility

Purpose: decide whether it is safe to generate prose.

Input:

- enriched SourceLedger;
- context warnings;

- evidence availability.

Processing:

- deterministic quality gate classifies the run as feasible, feasible with constraints, needs research, needs human decision, or infeasible.

Output:

- FeasibilityReport;
- if blocked: `complete.status = blocked`, `finalDraft = null`, no local fallback.

Role/model handoff:

- no LLM role is used;
- this deterministic gate decides whether later roles are allowed to write prose;
- if blocked, `strategy`, `writer`, and `review` are not called.

Trace: feasibility step.

4.5 postContract

Purpose: lock what the post must preserve.

Input:

- approved brief;
- context summary;
- enriched SourceLedger;
- feasibility result;
- publication-size profile and fabula size intent.

Processing:

- PostContractBuilder resolves title, thesis, audience, value, goal, CTA, platform/date/time, topic/fabula ids, allowed claim ids, qualified claims, forbidden moves, evidence obligations, fabula obligations, risks, and PublicationSizeContract.

Output:

- provider-free PostContract.

Role/model handoff:

- no LLM role is used;
- this step creates the invariant contract that all later roles must preserve;
- strategy may plan within the contract, writer may execute it, and review evaluates candidates against it.

Trace: postContract step.

4.6 rulePack

Purpose: turn the contract and editorial rules into machine-readable rules.

Input:

- enriched context artifact;
- SourceLedger;
- FeasibilityReport;
- PostContract;
- topic/fabula/publisher rules.

Processing:

- rulePack is marked running before provider-backed evidence interpretation, so a long JSON retry path is visible as rulePack, not as stale postContract;

- RuleRegistrySnapshot is compiled first;
- compatibility RulePack is derived from the registry;
- the provider request is compacted before JSON generation: full ruleRegistrySnapshot

stays in the artifact, while the model receives relevant hard/critical/evidence/ style/attribution/topic/fabula rules, accepted public evidence summaries, external ledger claims, evidence synthesis, and the locked contract;

- each provider attempt runs inside DRAFT_EVIDENCE_INTERPRETATION_TIMEOUT_SECONDS

so a stuck strategy-model call can fail, write a child AiRun, and continue to repair/backup/fallback instead of leaving rulePack running forever;

- EvidenceInterpretationService converts accepted internal/external evidence into

editorial implications, tensions, examples, limits, forbidden overclaims, reader value hooks, and rejected evidence uses before material planning;

- EvidenceInterpretationFidelityPolicy classifies the interpretation as

sufficient, partial, weak, or missing. Retry/repair and backup acceptance are recovery signals, not evidence degradation by themselves. Deterministic fallback always lowers evidence fidelity, and missing accepted evidence blocks a trusted editorial verdict.

Output:

- ruleRegistrySnapshot;
- compatibility rule-pack fields;
- evidenceInterpretation;
- evidenceInterpretationFidelity;
- attempts[] for the interpretation provider/repair/backup path;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- rule compilation itself uses no LLM role;
- evidence interpretation uses the strategy role through DRAFT_STRATEGY_MODEL, then default, repair, optional backup, and finally deterministic fallback;
- this step translates editorial obligations into machine-readable rule ids and then

translates evidence into usable editorial meaning;

- later role prompts receive rule ids and validator bindings so they do not work

from anonymous prose constraints only;

- material planning and writing receive interpretation artifacts instead of raw

citation snippets as their main source context.

Trace:

- rulePack step;

- child `AiRun.requestPayload.draftRunStep` = `evidenceInterpretation` when provider interpretation runs;

- `artifactPayload.progress.operations[]` contains `evidenceInterpretation` primary/repair/backup attempts while the step is running;

- child `AiRun.requestPayload.inputStats` records original/compact rule counts, claim counts, evidence counts, prompt char estimate, selected model, and timeout seconds;

- attempt records may include `status=timeout` plus operation timing notes such as prompt build, provider request, JSON/shape validation, and `AiRun` persistence;

- `rulePack.evidenceInterpretation` readable section in `/ai-runs?runId=...`

4.7 materialPlan

Purpose: decide what material is usable before strategy and writing.

Input:

- context summary;
- RulePack and RuleRegistry;
- EvidenceInterpretation;
- enriched SourceLedger;
- PostContract;
- strategy ContextPack.

Processing:

- material planner receives `usableEvidenceCandidates`;
- the candidate list is capped by `DraftRunBudget.maxUsableEvidenceCandidates`;
- material planner receives `evidenceInterpretation` and should prefer implications, usable examples, limits, and forbidden overclaims over raw public snippets;
- it must either choose usable evidence or explain rejection reasons;
- accountability guard rejects empty/unexplained evidence selection;
- retry sequence: primary -> repair -> optional backup model -> emergency deterministic fallback.

Output:

- MaterialPlan;
- availableEvidence;

- rejectedEvidence;
- claimsRequiringAttribution;
- qualifiedClaims;
- attempts[];
- accountability metadata;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL, then default, then repair/backup

where applicable;

- the role receives the strategy ContextPack, usable evidence candidates,

ledger claim ids, contract, and registry;

- it must return selected or rejected evidence with reasons;
- strategy and rhetoricalPlans consume the material plan instead of scanning the whole ledger again.

Trace:

- materialPlan step;
- child AiRun.requestPayload.draftRunStep = materialPlan.

4.8 strategy

Purpose: define the main strategy for the post.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- strategy ContextPack.

Processing:

- OpenRouter JSON call or deterministic fallback builds the draft strategy.

Output:

- DraftStrategy: thesis angle, opening move, argument sequence, fabula usage, CTA plan, forbidden moves, tone notes;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL or default;

- the role receives MaterialPlan plus the strategy ContextPack;
- it creates the main route of argument, but cannot change the PostContract;
- rhetoricalPlans expands this strategy into several executable routes.

Trace: strategy step and child AiRun.requestPayload.draftRunStep = strategy.

4.9 rhetoricalPlans

Purpose: build several routes for executing the same contract.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy.

Processing:

- one JSON call returns 2-3 rhetorical plans;
- retry sequence: primary -> repair -> optional backup -> deterministic fallback.

Output:

- RhetoricalPlanSet;
- plan moves, claim ids to use/avoid, required rule ids, CTA route, risks;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL, then repair/backup if JSON shape

is invalid;

- the role receives contract, material plan, draft strategy, claim ids, rule ids,

and the strategy ContextPack;

- it returns alternative rhetorical plans that the writer must execute;
- the writer must not invent a new route outside these plan artifacts.

Trace:

- rhetoricalPlans step;
- child AiRun.requestPayload.draftRunStep = rhetoricalPlans.

4.10 draft

Purpose: generate candidate drafts from rhetorical plans.

Input:

- request brief;

- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy;
- RhetoricalPlanSet;
- EvidenceInterpretation;
- writer ContextPack.

Processing:

- one candidate is generated per rhetorical plan;
- the executed candidate count is capped by `DraftRunBudget.maxDraftCandidates` before provider calls are attempted;
- each provider failure falls back per candidate, not for the whole step;
- candidate publishability guard marks candidates eligible, penalized, or excluded;
- deterministic v1 selection creates an initial scorecard.

Output:

- candidates[];
- selection;
- scorecard;
- child AiRun ids;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: writer;
- model resolver chooses `DRAFT_WRITER_MODEL` or default;
- each candidate call reloads persisted artifacts through `DraftRunContextAccessService`

and receives one `WriterDossierFactory` projection: rhetorical route, compact planning decisions, evidence/claim/rule handles, forbidden moves, and size/style contract;

- the standard writer projection retains one route, four curated evidence items, two claim handles, and five relevant rules; omitted optional detail stays trace-visible through dossier counts and handles rather than entering the provider prompt;

- each JSON candidate attempt follows the universal retry sequence: primary writer model, repair prompt, optional backup model, then domain-safe deterministic fallback only if provider attempts are exhausted;

- primary writer attempts use `DRAFT_WRITER_TEMPERATURE` and `DRAFT_WRITER_TOP_P`;
- repair and backup attempts use `DRAFT_JSON_REPAIR_TEMPERATURE`, not the creative

writer temperature;

- writer prompts forbid leaking internal artifact names such as `SourceLedger`, `publicEvidence`, `validators`, `RuleRegistry`, or `PostContract` as unexplained public dev-jargon;

- candidates write prose, but selection still remains downstream responsibility;

- review receives all candidates plus validation context.

Trace:

- draft step;
- child `AiRun.requestPayload.draftRunStep = draftCandidate;`
- direct `providerInput`, `providerDossier.runtimeMigrated=true`, `payloadBudget`, `inputStats`, and `payloadStats` for every primary/repair/backup attempt.

AS IS behavior:

- writer prompts now receive interpreted implications, examples, limits, and forbidden overclaims, so public evidence should shape the argument instead of being pasted as a decorative citation block;
- each provider attempt is visible through child `AiRun` audit with model role, selected model, attempt label, repair/backup marker, generation params, and safe validation error.

4.11 validation

Purpose: validate candidates, rank them, revise the selected candidate, and choose the final draft.

Input:

- draft artifact;
- context artifact;
- `RulePack` and `RuleRegistry`;
- `MaterialPlan`;
- review/writer context packs through child services.

Processing:

1. deterministic lint checks all candidates;
2. LLM validation checks all candidates and separates actionable findings from observations;
3. editorial critique challenges all candidates for idea strength, tension, author stance, source integration, generic prose, and reader value;
4. alternative-angle tournament may add one critic-driven challenger candidate; both route JSON and challenger-prose JSON use primary/repair/backup attempts, and no fake deterministic challenger is invented when all attempts fail;
5. final validation report is extended when a challenger enters the pool: initial candidate reports are reused, provider-heavy validation runs only for the new challenger, and the reports are merged into one final candidate-pool report;
6. pairwise ranking chooses the best candidate from the merged candidate pool;
7. deep revision loop builds validator and editorial goals, revises, re-runs deterministic validation, compares previous best vs revised across editorial dimensions, and accepts only targeted improvement without regression. Accepted cycles record concrete reasons such as resolved goals, clear pairwise win, and no deterministic regression;
8. final quality gate builds a `FinalQualityContract` from Fabula research depth,

publication kind, post contract, rule registry, and material plan. It checks the delivered candidate with deterministic public-prose rules and an independent finalGate model review. If it leaks internal pipeline names, reads like a source dump, loses author stance relative to the contract, or misses reader value, the gate creates bounded final repair instructions for the writer. The number of final repair cycles is controlled by DRAFT_FINAL_REPAIR_MAX_ITERATIONS (default 2). A repaired draft is accepted only when deterministic regression and gate findings improve. Attribution warnings are actionable only when they resolve to source-backed claim ids with expected markers that are missing from the final body. Free-text attribution requirements that cannot be resolved become diagnostic handoff noise; if the independent final-gate review passes source integration, that diagnostic noise does not launch final repair;

9. validation and final-gate warnings/critical findings are folded into

qualityFidelity.issueLifecycle. Each item is classified as resolved, suppressed, acceptedRisk, or open. Accepted final repair resolves the gate issue it repaired. Explicit suppression or accepted risk remains visible in trace. Open critical findings block trusted editorial quality, and an unresolved final-gate warning prevents cleanSuccess even when DraftRun.status is succeeded;

10. provider-heavy validation sub-operations persist progress as they run; if a late

operation fails, the operation is marked failed, safe error details are saved, and the previous best candidate is kept when one exists.

Output:

- deterministic validation report;
- llmValidationReport;
- editorialCritiqueReport;
- alternativeAngleTournament;
- rankingRevision;
- revisionLoop with editorial goals, dimension scores, rejected moves, and stop

reason;

- finalQualityGate with final quality contract, independent review attempts,

final-draft status, public-prose status, source-dump risk, actionable attribution findings, diagnostic attribution noise, attribution review closure, final repair goals, repair cycles, repair result, and gate-level final decision;

- qualityFidelity.issueLifecycle with lifecycle status and reason for validation

and final-gate warning/critical issues, including synthetic final-gate items when the gate status is warning or critical;

- progress with nested validation operations, child AiRun ids, failed-operation

status, and safe error details when a provider-heavy sub-operation fails;

- final draft candidate decision;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- review for LLM validation and pairwise ranking;
- critic for report-only editorial critique;
- anotherAngle for challenger route generation;
- finalGate for independent final public-prose acceptance;
- writer for directed revision and final public-prose repair.
- deterministic lint runs first without a model;

- review receives candidates, review ContextPack, deterministic findings, rules, ledger, and contract, then returns report-only findings and pairwise ranking;
- critic receives candidates, critic ContextPack, evidence interpretation, and validation context, then returns report-only critique findings and observations;
- anotherAngle receives an AlternativeAngleDossierFactory projection with compact candidate summaries, critique signals, validation issues, rejected moves, contract invariants, and supporting handles, then returns one different contract-safe route;
- route input is bounded to three candidate summaries, three critique summaries, three validation issues, two rejected moves, two evidence items, and two rules; each retained rejected-move handle resolves through the persisted editorial critique;
- the accepted route is persisted into the running validation artifact before WriterDossierFactory(alternativeAngleCandidate) resolves it for challenger prose;
- finalGate receives the delivered final candidate, FinalQualityContract, deterministic gate findings, validation context, evidence interpretation, ledger, rule registry, material plan, and attribution requirement resolution, then returns independent findings, observations, and final repair goals;
- writer may execute bounded final public-prose repair cycles after the revision loop if finalQualityGate returns warning or critical;
- writer receives the current best candidate plus validator repair goals, editorial goals, rejected moves, dossier/context pack, and constraints, then returns revised prose;
- directed revision attempts use DRAFT_REVISION_TEMPERATURE and DRAFT_REVISION_TOP_P; malformed-JSON repair and backup attempts use the JSON repair temperature;
- regression guard checks deterministic health, and review compares old-vs-new by idea strength, tension, reader value, author stance, source integration, structure, and validator health;
- acceptance policy decides whether the revised prose replaces the current best;
- complete receives only the final decision, not the whole model conversation.

Trace:

- validation step;
- child AiRun.requestPayload.draftRunStep = llmValidation;
- child AiRun.requestPayload.draftRunStep = editorialCritique;
- child AiRun.requestPayload.draftRunStep = alternativeAngleRoute;
- child AiRun.requestPayload.draftRunStep = alternativeAngleCandidate;
- child AiRun.requestPayload.draftRunStep = pairwiseRanking;
- child AiRun.requestPayload.draftRunStep = directedRevision.

4.12 complete

Purpose: mark the run result.

Output cases:

- normal success: parent DraftRun has finalDraft;
- quality-blocked success: parent DraftRun has finalDraft = null and complete.status = blocked;

- failure: parent DraftRun status is failed with a safe error.

Role/model handoff:

- no LLM role is used;
- this step only records the final orchestration outcome for the frontend and trace workbench.

4.13 Post-run editor version loop

Purpose: let the human editor keep improving or manually editing the delivered machine draft without mutating older versions.

Inputs:

- parent DraftRun.finalDraft, stored locally as DraftVersion v1 with source=machineFinal;
- active draft version chosen by the editor;
- optional editor comment;
- optional compact machine trace context from the parent DraftRun: final quality

gate summary, revision loop summary, alternative-angle outcome, validation summary, unresolved risks, post contract, and material evidence.

Behavior:

- the editor can switch between saved versions;
- manual text edits are local until the user saves them as a new source>manualEdit version;
- one click on Улучшить по комментарию calls POST /api/drafts/revise-with-comment;
- that endpoint is not a new DraftRun. It is a post-run writer-role JSON call with child AiRun audit and existing JSON retry discipline;
- after a successful writer revision, the endpoint runs a lightweight review role quality check. It evaluates whether the revised version followed the editor comment, preserved visible source markers from the base version, avoided internal pipeline jargon, and did not regress as public prose;
- quality check failure or provider unavailability does not cancel the created version. The version receives qualityCheck.status = notRun with attempt metadata;
- provider failure does not create a fake version;
- the editor can mark any saved version as final, including v1 after later versions exist;
- final version selection creates or updates one deterministic editorialLearning note in Author Memory. The note is visible immediately with status pendingReview, but it does not influence author-position inference until the editor accepts it.

Output:

- local PostDraft.versions[], activeVersionId, and finalVersionId;

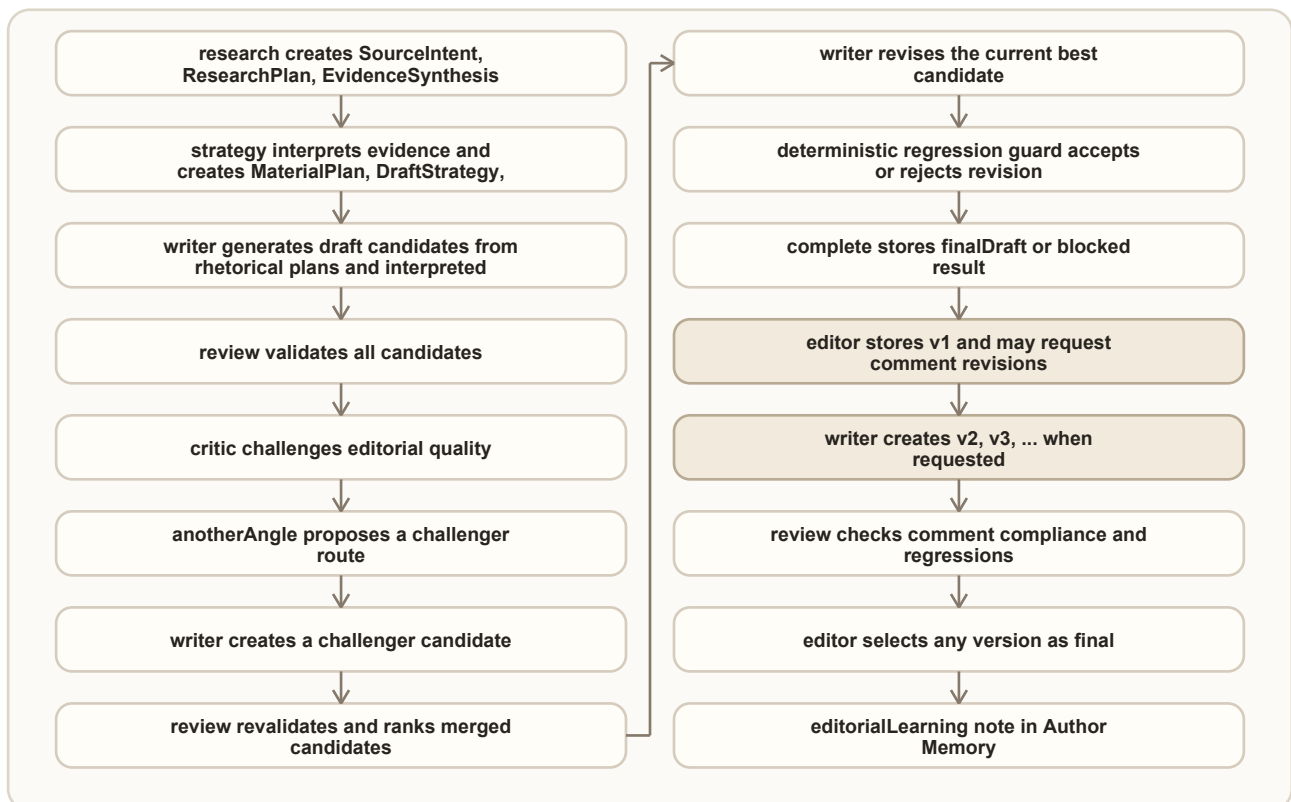
- compatibility mirror fields `PostDraft.title/body/version` always reflect the active version;
- `FinalText.draftVersionId`, `FinalText.versionNumber`, and `FinalText.editorDecisionSnapshot`;
- `DraftVersion.qualityCheck` for human-comment revisions when the review ran or explicitly could not run;
- one `AuthorNote.type = editorialLearning` with selected/rejected version metadata, comment summaries, manual edit count, quality-check summaries, unresolved risks, suggested takeaway, and status `pendingReview | accepted | rejected`.

Role/model handoff:

- post-run human-comment revision uses the writer role;
- it receives a compact machine context, not raw `DraftRun JSON`;
- it must preserve source markers, allowed claims, forbidden moves, and final quality constraints.
- post-run human revision quality check uses the review role;
- it is diagnostic: it can mark the version as passed, warning, critical, or not-run, but it cannot block saving or final approval.

5. Role model usage and handoff AS IS

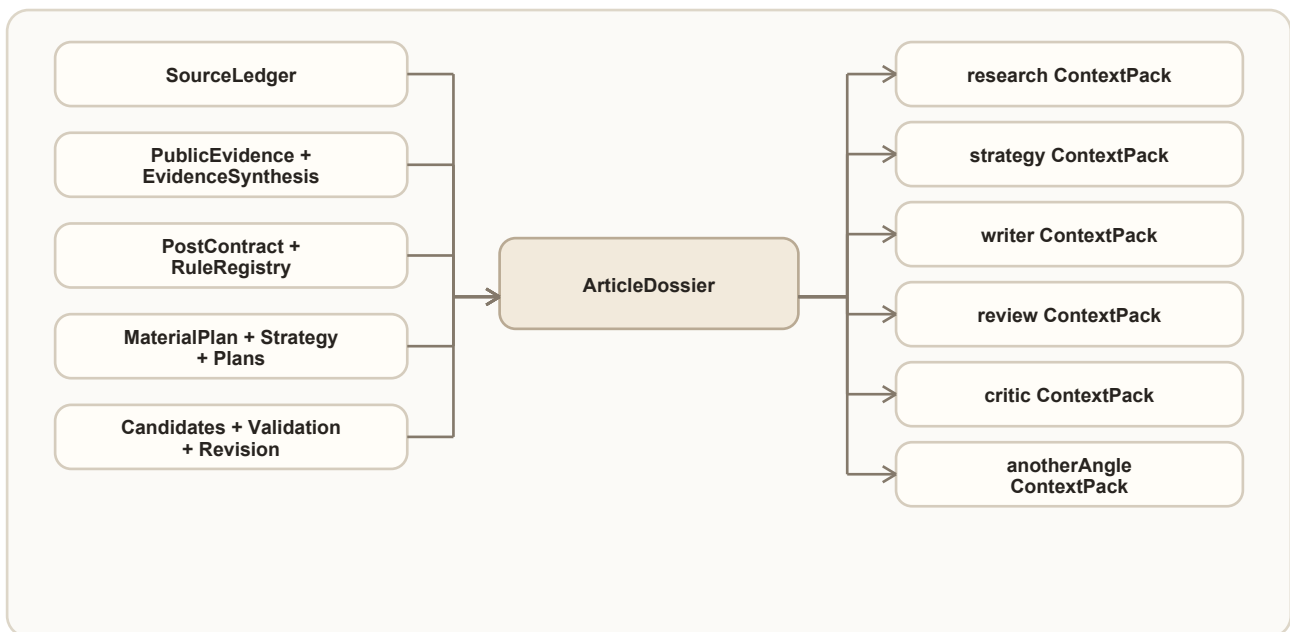
The current role interaction is artifact-mediated. A role does not continue the previous model's hidden conversation. It receives explicit artifacts and a role-specific context pack, then writes a new artifact for the next role.



Role summary:

Role	Used now	Receives	Must not do
research	source research planning and evidence synthesis	source intent, public evidence, context	invent facts or strengthen weak evidence
strategy	evidence interpretation, material plan, draft strategy, rhetorical plans	strategy context pack, rules, contract, material, evidence synthesis	change post thesis or bypass contract
writer	draft candidates and directed revisions	writer context pack, evidence interpretation, material plan, rhetorical plan, repair goals	create new claims or violate forbidden moves
review	LLM validation and pairwise ranking	review context pack, candidates, rules, ledger, validation reports	rewrite prose directly
critic	report-only editorial critique	critic context pack, evidence interpretation, candidates, validation context	rewrite prose or change final selection
anotherAngle	one critic-driven challenger route	another-angle context pack, initial validation, editorial critique, rejected/weak moves	act as technical backup or invent new evidence
writer post-run	human-comment revision of one saved version	active draft version, editor comment, compact machine trace context	mutate prior versions or fabricate a successful revision after provider failure
review post-run	comment-compliance quality check for human-comment revisions	base version, revised version, editor comment, compact machine trace context	block saving a successful revision or turn warnings into learning signals automatically

6. Context flow



Important AS IS rules:

- raw DraftRun trace is not the normal prompt context;
- ArticleDossier is rebuilt from selected artifacts;
- ContextPack selects up to 12 cards for one role;
- child AiRun.requestPayload stores the role model and context pack it received;
- context packs are local orchestration artifacts, not long-term memory.

Slice 2.17.4.6.1.3.6 adds an available provider-free context/dossier foundation:

- DraftRunContextAccessService reads compact contract, evidence, rule, planning, candidate, validation, and final-quality views deterministically;
- planning, writer, review, ranking, revision, and final-quality dossier factories return typed ProviderDossier contracts;
- compact entries carry stable handles back to full persisted artifacts;
- missing required inputs block readiness, while trimming and missing optional input remain explicit quality risk;
- scripts/audit_draft_run_provider_dossiers.py replays all six dossier families without providers and reports actual projected size, unresolved handles, and forbidden-field violations.

The foundation is now connected across the target runtime operations. Planning operations materialPlan, strategy, rhetoricalPlans, draftCandidate, alternativeAngleRoute, alternativeAngleCandidate, llmValidation, pairwiseRanking, directedRevision, and finalQualityGateReview use typed dossiers. The 3.9 code path is current AS IS and live-accepted. Run 7bf3... proves all four target operation families inside their 22000/24000 caps, sufficient evidence, publishable final quality, and zero open critical/warning issues.

6.1 Context Handoff and Provider Input Contract

This section defines what future DoD must check before changing provider-heavy DraftRun behavior.

Terms:

- rich artifact: the full persisted DraftRun artifact. It stays in storage and trace for audit/debug.
- provider projection: the curated subset sent to one provider call. It may be a role ContextPack, a payload-budget compact payload, an operation-specific dossier, or a transitional compact structure.
- direct budget proof: the current provider call itself records payloadBudget limits, sent/trimmed counts, prompt estimate, token estimate, and suppressed fields in child AiRun.requestPayload, attempts, or the current operationEnvelope.
- trace proof: the concrete trace path that proves what was sent, what was trimmed, which model was used, and how retry/backup/fallback behaved.
- known debt: an AS IS limitation that is deliberately carried forward and linked to a repair slice. Known debt is not target architecture.

Strict AS IS invariants:

- the step order remains
context -> sourceIntent -> publicEvidence -> feasibility -> postContract -> rulePack -> materialPlan -> strategy -> rhetoricalPlans -> draft -> validation -> complete;
- roles communicate through persisted artifacts, not hidden model conversation state;
- DraftRun.status=succeeded means the worker completed, not that the text is trusted editorial quality;
- provider retry, repair, backup, fallback, not-run, timeout, and schema/JSON incidents must be visible through structured trace;

- a nested payloadBudget copied from an older artifact is not direct proof that the current provider call was bounded.

Provider-heavy AS IS contract:

Operation	Current rich artifacts	Current provider projection / ContextPack	Required trace proof	Direct budget status	Known debt / target slice
sourceIntentAndResearchPlan	brief sources, context summary, initial source ledger, DraftRun budget	research prompt over brief sources, normalized context summary, initial ledger, budget caps	child AiRun, sourceIntent.researchPlan, skipped budget trace	partial/transitional; planning output has trace but no full provider-input dossier	Dossier boundary target after the planning budget gate
evidenceSynthesis	public evidence attempts, accepted/rejected evidence, initial/enriched ledger	accepted public evidence plus relevant ledger/context	publicEvidence.evidenceSynthesis, child AiRun, accepted/rejected evidence trace	partial/transitional	Dossier and evidence-query access target in 2.17.4.6.1.3.6
evidenceInterpretation	enriched ledger, accepted public evidence, evidence synthesis, contract, rule subset	compact evidence payload plus strategy ContextPack, timeout envelope	rulePack.evidenceInterpretation, evidenceInterpretationFidelity, child AiRun, operationEnvelope, timeout/input stats	enforced for representative operation	Further context-access hardening in 2.17.4.6.1.3.6
materialPlan	evidence interpretation, usable evidence candidates, post contract, rule registry, strategy context	PlanningDossierFactory(materialPlan) projection: compact post contract, interpreted evidence/claim handles, relevant rules, attempt repair context	materialPlan.materialPlan, selected/rejected evidence reasons, child AiRun.requestPayload.providerDossier.runtimeMigrated=true, direct providerInput, payloadBudget, inputStats, payloadStats	enforced per primary/repair/backup call; replay baseline projects 16810 chars with no forbidden fields	Runtime dossier migration complete in 2.17.4.6.1.3.7; live quality proof is stored with the slice evidence
strategy	material plan, post contract, rule registry, strategy context	PlanningDossierFactory(strategy) projection: compact post contract, MaterialPlan, interpreted evidence/handles, relevant rules, attempt repair context	strategy.draftStrategy, child AiRun.requestPayload.providerDossier.runtimeMigrated=true, direct providerInput, payloadBudget, inputStats, payloadStats, model role metadata	enforced per primary/repair/backup call; budget profile remains draftStrategy; replay baseline projects 17720 chars	Runtime dossier migration complete in 2.17.4.6.1.3.7
rhetoricalPlans	strategy, material plan, claim/rule ids	PlanningDossierFactory(rhetoricalPlans) projection: compact post contract, MaterialPlan, DraftStrategy, evidence/rule handles, attempt repair context	rhetoricalPlans.rhetoricalPlanSet, attempts, child AiRun.requestPayload.providerDossier.runtimeMigrated=true, direct providerInput, payloadBudget, inputStats, payloadStats	enforced per primary/repair/backup call; replay baseline projects 17203 chars	Runtime dossier migration complete in 2.17.4.6.1.3.7

Operation	Current rich artifacts	Current provider projection / ContextPack	Required trace proof	Direct budget status	Known debt / target slice
draftCandidate	one rhetorical plan, material evidence, contract, generation params	WriterDossierFactory(draftCandidate) projection with one route, compact planning, four evidence items, two claim handles, five rules, forbidden moves, and size/style constraints	draft.candidates[], child AiRun.requestPayload.providerDossier.runtimeMigrated=true, direct budget proof, fallback/publishability status	enforced per primary/repair/backupp call; standard cap 24000 chars	Runtime dossier migration complete in 2.17.4.6.1.3.8
llmValidation	one persisted candidate, deterministic findings, post contract, evidence/rule handles	ReviewDossierFactory projection with exact candidate prose, compact candidate-specific findings and bounded optional evidence/rules	validation.llmValidationReport, candidate reports, child AiRun.requestPayload.providerDossier, direct budget and incidents	enforced per attempt; standard cap 22000; accepted live maximum 16628	Runtime migration and live acceptance complete in 2.17.4.6.1.3.9
editorialCritique	candidate s, evidence interpretation, validation findings, critique context	critic ContextPack, candidate summaries, evidence interpretation, findings	validation.editorialCritiqueReport, child AiRun, operationEnvelope, parser/trace payload	enforced for representative operation	Debate/critique expansion remains future work; current single critique is AS IS
alternativeAngleRoute	persisted initial validation, editorial critique, rejected/weak moves, contract and candidate summaries	AlternativeAngleDossierFactory selection followed by AlternativeAngleRoutePayloadCompact: three symmetric candidate projections, candidate-scoped critique/validation signals, two rejected moves, compact contract, two evidence items, two rules, and bounded repair context	persisted tournament route before challenger construction, child AiRun with direct current-call budget, actual serialized messageCharCount, trimming counts, route attempts, resolvable editorial-critique handles	enforced per primary/repair/backupp call; standard cap 22000; live run 92532... records 16321 actual message chars with no incident	Runtime migration completed in 3.8; data-dependent cap drift repaired and live-accepted in 2.17.4.6.1.3.9.1
alternativeAngleCandidate	persisted challenger route, compact planning, contract/evidence constraints	WriterDossierFactory(alternativeAngleCandidate) projection with route, planning decisions, four evidence items, one claim handle, four rules and one critique summary	challenger candidate payload, child AiRun with direct budget proof, no-fake-challenger failure trace	enforced per primary/repair/backupp call; standard cap 24000 chars	Runtime dossier migration complete in 2.17.4.6.1.3.8
pairwiseRanking	persisted active candidate pool, candidate-scoped validation reports, post contract and evidence handles	RankingDossierFactory projection with equal opening/middle/ending windows, per-candidate finding counts/summaries, seven dimensions and selection constraints	validation.rankingRevision.pairwiseRanking, attempts, scorecard, child direct budget	enforced per attempt; standard cap 22000; accepted live maximum 20534	Runtime migration and live acceptance complete in 3.9; live comparisons preserve winner and seven dimensions but blank left/right pair ids remain explicit trace debt 3.9.2

Operation	Current rich artifacts	Current provider projection / ContextPack	Required trace proof	Direct budget status	Known debt / target slice
directedRevision	current persisted candidate , applicable repair goals, anti-regression constraints and evidence/ rule handles	RevisionDossierFactory projection with exact candidate body and bounded prioritized repair context	validation.rankingRevision.revisionLoop, child AiRun.requestPayload.providerDossier, accepted/rejected cycles and direct budget	enforced per attempt; standard cap 24000; accepted live maximum 20157	Runtime migration and live acceptance complete in 2.17.4.6.1.3.9
finalQualityGateReview	exact candidate being delivered or tested after repair, final-quality contract, deterministic gate, applicable findings and repair history	FinalQualityDossierFactory projection; profile finalQualityReviewRepair records operationAlias=finalQualityGateReview	effective final gate, issue lifecycle, child direct budget, repair cycles and regression decision	enforced per attempt; standard cap 22000; accepted live maximum 18015	Runtime migration and live quality acceptance complete in 2.17.4.6.1.3.9
humanCommentRevision	active DraftVersion, editor comment, compact machine trace context	compact current/base version, comment, contract/material evidence and machine trace summaries	new version result, child AiRun, operationEnvelope, attempts, safe error	enforced for representative operation	Current HITL context is already compact; future learning slices may tighten accepted-risk handling
humanCommentRevisionQualityCheck	base version, revised version, editor comment, compact machine trace context	compact base/revised versions and quality constraints	DraftVersion.qualityCheck, child AiRun, operationEnvelope, not-run/incident trace	enforced for representative operation	Current HITL check is diagnostic and must not block saving the revision

Future provider-input DoD should use this table mechanically: a change is not ready unless it proves the relevant row still has the required trace proof, or updates the row and links the new debt/repair slice.

7. Artifact catalog

Artifact	Created by	Read by	Debug location
SourceLedger	context/publicEvidence	feasibility, contract, rules, planning, validation	context.sourceLedger, publicEvidence.enrichedSourceLedger
SourceIntent	sourceIntent	publicEvidence	sourceIntent step
ResearchPlan	sourceIntent	publicEvidence	sourceIntent.researchPlan
PublicEvidenceBatch	publicEvidence	evidence synthesis, trace	publicEvidence.items, publicEvidence.attempts

Artifact	Created by	Read by	Debug location
EvidenceSynthesis	publicEvidence	ledger merger, material planning, trace	publicEvidence.evidenceSynthesis
DraftRunBudget	context	sourceIntent, publicEvidence, material plan, draft, validation	context.draftRunBudget, step budget metadata
EvidenceInterpretation	rulePack	material plan, writer, dossier, trace	rulePack.evidenceInterpretation
PostContract	postContract	rules, planning, validation, revision	postContract step
RuleRegistrySnapshot	rulePack	material plan, validation, revision	rulePack.ruleRegistrySnapshot
MaterialPlan	materialPlan	strategy, writer, validation, ranking	materialPlan.materialPlan
DraftStrategy	strategy	rhetorical plans, writer	strategy.draftStrategy
RhetoricalPlanSet	rhetoricalPlans	draft candidates	rhetoricalPlans.rhetoricalPlanSet
DraftCandidate	draft	validation, ranking, revision	draft.candidates[]
ValidationReport	validation	ranking/revision, trace	validation.candidateReports
LlmValidationReport	validation	ranking/revision, trace	validation.llmValidationReport
EditorialCritiqueReport	validation	alternative-angle tournament, trace, dossier	validation.editorialCritiqueReport
AlternativeAngleTournament	validation	final validation/ranking, trace	validation.alternativeAngleTournament
RankingRevision	validation	final draft selection, trace	validation.rankingRevision
FinalQualityGate	validation	final draft selection, trace	validation.rankingRevision.finalQualityGate
QualityFidelityReport	workflow completion	diagnostics, editor-facing risk summary, future reliability analytics	validation.rankingRevision.qualityFidelity, complete.qualityFidelity
ArticleDossier	article memory service	context pack builder, trace	selected step artifacts
ContextPack	article memory service	child LLM services	step artifacts and child AiRun.requestPayload
DraftVersion	frontend/editor actions	editor version list, final selection	local workspace postDraft.versions[]
HumanCommentRevisionQualityCheck	post-run revise-with-comment endpoint	editor version list, final selection, future learning slice	local workspace postDraft.versions[].qualityCheck, child AiRun.requestPayload.draftRunStep = humanCommentRevisionQualityCheck
EditorDecisionSnapshot	final text approval	editorial-learning note builder, audit/debug	local workspace finalText.editorDecisionSnapshot
EditorialLearningAuthorNote	final text approval	author memory feed, optional author-position inference after acceptance	local workspace authorNotes[], type editorialLearning

8. Retry, fallback, and blocked behavior

Area	Behavior
live DraftRun	frontend keeps polling; no compatibility fallback while run is alive
queued DraftRun	queue-wait state; old updated_at alone is not a generic stale signal while status is queued

Area	Behavior
stale DraftRun	diagnostic state after no timestamp progress or after a current provider operation exceeds its stale budget; not automatic fallback
slow-but-healthy provider operation	running step progress has currentOperationId, currentOperationStartedAt, selectedModel, and operation age inside staleAfterSeconds; do not classify as unexplained hang
over-budget provider operation	stale diagnostics report the current provider operation, model, wait seconds, and stale budget instead of the old generic five-minute message
slow-but-healthy validation	validation has runtimeBudget.currentOperationId and heartbeat/current-operation age inside runtimeBudget.limits.staleAfterSeconds; do not classify as stuck
over-budget validation operation	stale diagnostics report the current validation operation and runtime stale budget instead of the old generic five-minute message
public search disabled	attempt is notConfigured; it is not proof
malformed JSON planning	repair retry, optional backup model, then deterministic fallback
malformed, empty, or timed-out evidence interpretation	failed child AiRun, failed nested operation, repair retry, optional backup model, then deterministic interpretation fallback
editorial critique provider missing	editorialCritiqueReport.status=not-run; no fake critique
malformed editorial critique JSON	repair retry, optional backup model, then not-run for that candidate
alternative-angle provider missing	alternativeAngleTournament.status=not-run; original candidates continue
malformed alternative-angle route JSON	repair retry, optional backup model, then tournament failed; no deterministic fake angle
alternative-angle candidate provider failure	repair retry, optional backup model, then tournament failed; original candidates continue
validation sub-operation failure	mark nested operation failed, preserve partial artifact, keep previous best if available
material plan ignores evidence	accountability retry, optional backup model, then emergency fallback
unresolved attribution requirement	record diagnostic metadata; not a repair goal unless it resolves to expected source markers that are missing from the final body
human-comment revision provider failure	no new version is created; UI keeps the current version list and shows a safe error
human revision quality-check failure	created version is kept; qualityCheck.status=notRun records attempts and safe reason
candidate provider failure	fallback only for that candidate direction
fallback candidate selection	fallback is diagnostic unless publishability guard allows it
feasibility blocked	run succeeds with finalDraft = null; no local fallback
all invalid candidates	run succeeds as quality-blocked; no local fallback
revision regression	keep previous best candidate

Slice 2.17.4.6.0.3.2 adds shared operationEnvelope governance for representative provider-heavy operations. Migrated operations expose operationId, operation kind, owner, status, attempts, child AiRun ids, input/payload stats, retry policy, timeout profile where applicable, incident metadata, safe error, and result payload. Fallback, not-run, failed, timeout, cancelled, and stale outcomes require incident metadata. Backup success keeps the accepted payload but records backupAccepted. Representative migrated operations are evidenceInterpretation, editorialCritique, directedRevision, humanCommentRevision, and humanCommentRevisionQualityCheck; other current provider-heavy operations remain in CURRENT_LLM_OPERATION_INVENTORY until their owning migration slices.

Slice 2.17.4.6.0.3.3 adds DraftRun provider-input payload budget governance. Full artifacts stay in parent DraftRun storage, but enforced provider calls receive a curated compact payload immediately before prompt messages are built. The enforced representative operations are evidenceInterpretation, editorialCritique, directedRevision, humanCommentRevision, and humanCommentRevisionQualityCheck. Each records

payloadBudget.profileId, execution mode, limits, sent/trimmed counts, suppressed fields, semantic input contract, qualityRisk, promptCharEstimate, and approxTokenEstimate in child AiRun.requestPayload, attempts, and operationEnvelope.payloadStats. If compacted input still exceeds the profile cap, trace records contextOverBudget; hard-cap breaches record payloadTooLarge. Remaining provider-heavy operations are explicit payload-budget debt entries in CURRENT_LLM_OPERATION_INVENTORY.

Slice 2.17.4.6.0.3.4 adds ValidationRuntimeBudget governance for the validation stage. The runtime guard records runtimeBudget.profileId, execution mode, limits, used counters, startedAt, lastHeartbeatAt, currentOperationId, currentOperationStartedAt, slowButHealthy, stopReason, exhausted, and incidents in the existing validation progress artifact. It uses canonical stop reasons: acceptedQuality, humanReviewRequired, budgetExhausted, maxIterations, noImprovement, and providerIncident; older internal reasons are kept as detailStopReason where useful. No endpoint contract or SQLite schema is changed.

Slice 2.17.4.6.1.3.4 adds ProviderOperationRuntime diagnostics for non-validation provider-heavy steps that previously only showed the parent step as running. materialPlan, strategy, and rhetoricalPlans now write a current operation into the existing step progress payload before the provider call starts. The progress payload includes currentOperationId, currentOperationStartedAt, operationKind, modelRole, selectedModel, promptCharEstimate, approxTokenEstimate, staleAfterSeconds, providerWaitSeconds, and slowButHealthy when available.

Slice 2.17.4.6.1.3.5 adds direct provider-input budget enforcement for the planning trio that produced the largest live prompt estimates: materialPlan, strategy, and rhetoricalPlans. These calls now cross ProviderInputBudgetGate before build_*_messages(...) and persist current-call proof in child AiRun.requestPayload: operationId, providerInput, payloadBudget, inputStats, and payloadStats. The gate uses the existing semantic contracts and payload compaction policy, so full parent DraftRun artifacts remain available for diagnostics while provider messages receive the compact projection. The strategy pipeline trace uses operationId=strategy; its existing budget profile remains draftStrategy and is recorded as payloadBudget.profileId=draftStrategy plus payloadBudget.operationAlias=strategy.

Fresh live proof for this slice shows the current transition state clearly: materialPlan, strategy, and rhetoricalPlans now write direct current-call budget proof, but their compacted inputs can still be classified as overBudget. That is not a clean provider-input verdict. It means the call crossed the gate and the oversized context is visible before/during the provider attempt, while actual prompt-size reduction remains the responsibility of the planning dossier migration slice 2.17.4.6.1.3.7.

Slice 2.17.4.6.1.3.7 closes that planning transition. Before each planning step, LegacyDraftRunPhaseBuilder reloads the persisted DraftRun and uses DraftRunContextAccessService plus PlanningDossierFactory to assemble exactly one operation-specific dossier. Prompt builders receive only its budgeted projection and attempt-specific repair context. Full rulePack, SourceLedger, ArticleDossier, ContextPack, previous operation envelopes, and previous budget objects are forbidden from provider input. The full parent artifacts remain stored for replay, deterministic fallback, and accountability. Dossier replay over the control run reports zero unresolved handles and zero forbidden-field violations. Runtime migration and strict live acceptance were completed through Slice 3.9. Run 7bf3a7b9-4646-4b32-8bfd-cc5b200a1b47 records 16/16 target attempts directly budgeted, accepted maximums of 16628/20534/20157/18015 characters, sufficient evidence, editorial status publishable, and open critical/warning counts 0/0.

Slice 2.17.4.6.1.3.9.1 closes the remaining alternativeAngleRoute cap drift. The operation now budgets its dossier and repair context together, uses equal candidate projections, and verifies the final serialized messages rather than only the nested provider-input JSON. Live run 92532cb9-e83b-4bb1-ab2b-7d7a46d279b5 records 14371 provider-input characters and 16321/22000 actual message characters with no budget incident. The challenger was generated, validated, and became the basis of the final revised winner; evidence remained sufficient, editorial status publishable, and open critical/warning counts 0/0. The preceding attempt d08b26c7-2b5d-436a-8d03-5f4def3b3991 ended before route execution because of a transient SQLite disk I/O error; both databases passed integrity checks and that run is not used as route-quality proof.

Accepted live proof c2303e05-e7d0-4cad-a3f9-6ea26fc1a3ed records actual planning inputs of 14900 chars (materialPlan), 15017 chars (strategy), and 13601 chars (rhetoricalPlans) against historical baselines 207065, 210584, and 45249. All planning attempts are directlyBudgeted; there are no planning overBudget, payloadTooLarge, contextOverBudget, missing-direct-budget, unresolved-handle, or forbidden-field findings. The run completed with sufficient evidence coverage, trusted/publishable editorial status, and no open critical or warning issues. Provider malformed JSON caused a visible safe strategy fallback, but the dossier had no missing required inputs and no budget incident. The safe comparison is stored under docs/evidence/draft-runs/e874fd2b-cfa0-4b6a-815d-c0cf6d9763d2/.

The replayable provider-input audit is:

```
python scripts/audit_draft_run_provider_inputs.py --run-id <DraftRun ID> --format json
```

It reads stored child AiRun.requestPayload records and classifies each target operation as directlyBudgeted, overBudget, missingDirectBudget, nestedBudgetFalsePositive, or explicitDebt. A nested payloadBudget inside an older artifact is deliberately classified as nestedBudgetFalsePositive, not as proof of the current call. The target list includes directedRevision; no review, ranking, revision, or final-gate operation is exempt from direct proof.

Slice 2.17.4.6.1.3.9 additionally scopes issue lifecycle by delivered candidate. Findings for non-delivered candidates remain diagnostic and are resolved with candidate-not-delivered; only findings applicable to the effective final candidate contribute to open critical/warning counts. Historical unknown scope remains open. The detailed implementation and acceptance evidence is stored in COMPARISON_2_17_4_6_1_3_9.md. Fresh run 7bf3... closes the target-operation budget and quality proof; the unrelated alternativeAngleRoute cap drift is explicit debt 2.17.4.6.1.3.9.1.

During the live proof for 2.17.4.6.1.3.5, a long Docker DraftRun exposed a separate SQLite durability failure: var/glavred-draft-runs.sqlite3 became malformed after a worker disk I/O error during validation progress persistence. This is not treated as provider-input budget behavior. It is tracked as follow-up slice 2.17.4.6.1.3.5.1, because future live-heavy DraftRun slices need storage durability before their provider/runtime findings can be trusted.

Slice 2.17.4.6.1.3.5.1 adds the operational storage guard without changing the pipeline flow, public response contract, SQLite schema, prompts, provider behavior, or budget policy. SqliteDraftRunRepository and SqliteAiRunRepository now use backend.app.infrastructure.sqlite_runtime.SqliteConnectionFactory, which opens local SQLite connections with a bounded timeout, busy_timeout, journal_mode=WAL, synchronous=NORMAL, foreign_keys=ON, row factory, and explicit commit/rollback. If SQLite reports malformed/unavailable storage, API and diagnostics should classify the problem as storage durability, not provider quality, prompt quality, or budget-gate failure. The repeatable local check is:

```
python scripts/check_sqlite_integrity.py --format json --fail-on-error
```

If the check reports sqliteDatabaseMalformed, preserve the ignored var/ DB file as evidence, restore a clean working DB, rerun the integrity check, and only then use new DraftRun traces as provider/runtime proof. The live incident 89dca24d-c06e-4163-97db-0b59aaaf81b4 remains the reference example for this storage failure class.

Additional Docker/Windows live-proof caveat from Slice 2.17.4.6.1.3.8: direct host-side reads of var/glavred-ai-runs.sqlite3 while the Docker worker is actively writing can still expose or trigger a malformed local view of the bind-mounted DB. During an active Docker DraftRun, progress should be observed through the HTTP API and worker logs. Host-side SQLite diagnostics and provider-input audits should run after terminal status, or after stopping backend/worker. The incident DB was preserved as var/glavred-ai-runs.sqlite3.corrupt-20260710-221934; the repeat proof 72b3a2df-c8be-41a8-b2c0-e74e5e502cd0 finished with both draftRuns and aiRuns integrity checks returning ok.

The public DraftRun response shape is unchanged: `isStale`, `staleReason`, and `lastProgressAt` remain computed fields. Internally, queued runs are classified as `queue-wait` rather than generic `stale`, validation still uses `ValidationRuntimeBudget`, and diagnostics can now distinguish `queue wait`, `slow provider wait`, `validation runtime health`, and a genuinely `stale provider operation`.

Slice 2.17.4.6.1.0 adds `QualityFidelityReport` as a trace-only DraftRun artifact. It does not change step order, prompts, provider selection, API, or SQLite schema. The report separates `technicalStatus`, `providerRecoveryStatus`, `editorialStatus`, and `overallVerdict`. Successful primary repair/retry on the same model is normal recovery, not a quality failure. Backup success is accepted but diagnostic. Domain-safe deterministic fallback lowers fidelity. Step-level quality problems are tracked separately: evidence coverage, unresolved critical/warning findings, final-gate warning/critical status, rejected final repair, and size/over-budget risks.

Slice 2.17.4.6.1.3 adds `ProviderReliabilityReport` as a developer diagnostics artifact over multiple stored DraftRuns. It reads structured `qualityFidelity`, operation envelopes, payload/runtime budgets, and child `AiRun` records; it does not parse prose diagnostics and does not change runtime behavior. One run is reported as `insufficientData` for systemic conclusions. Every non-clean signal must become a remediation decision such as `noActionExpected`, `watchWithMoreRuns`, `fixBacklogSlice`, or `fixBeforeTrustingQuality`.

Slice 2.17.4.6.1.3.3 tightens provider JSON recovery diagnostics. Empty provider content, missing required keys, invalid response shape, and operation-specific contract mismatches are classified as `schemaFailure`; JSON parser failures remain `malformedJson`; `unknownProviderFailure` is reserved for failures without a more specific structured cause. Historical stage summaries with `unknownProviderFailure` may be refined from matching child `AiRun.error` records during replay. Isolated same-model repair and isolated fallback are watch signals; repeated fallback or quality-impact fallback remains repair work.

The report also includes `signalCoverage`. This is the audit layer for the analytics itself: it records raw child `AiRun` records, operation-envelope incidents, `retry/backlog/fallback` signals, `payload/runtime budget` incidents, and ignored stats-only budget payloads with reasons. `fixBacklogSlice` and `fixBeforeTrustingQuality` remediation decisions must point to concrete roadmap slices.

9. How to read a run trace

Open `/ai-runs?runId=<DraftRun ID>` and inspect in this order:

1. `context`: confirm source signal, candidate, topic, fabula, rules are present.
2. `DraftRun budget`: confirm research depth, execution mode, effective caps, used counts, cap hits, skipped tasks, and trimmed evidence/claims.
3. `sourceIntent`: confirm research requests were normalized correctly.
4. `publicEvidence`: confirm URL/search attempts, accepted evidence, rejected drift, evidence synthesis, and enriched ledger.
5. `feasibility`: confirm run was allowed or blocked for quality reasons.
6. `postContract`: confirm thesis, CTA, size, allowed claims, and forbidden moves.
7. `rulePack`: confirm rule registry exists and size/evidence rules are present.
8. `evidenceInterpretation`: inside `rulePack`, confirm implications, examples, limits, forbidden overclaims, rejected evidence uses, attempts, `operationEnvelope`, timeout profile, input stats, `payloadBudget`, and incident taxonomy.
9. `materialPlan`: confirm selected evidence and explicit rejection reasons.
10. `materialPlan / strategy / rhetoricalPlans progress`: while running, inspect

progress.currentOperationId, currentOperationStartedAt, selectedModel, prompt/token estimates, providerWaitSeconds, staleAfterSeconds, and slowButHealthy before calling the run stuck.

11. strategy: confirm the strategy does not change the contract.

12. rhetoricalPlans: confirm the plans are different routes, not new topics.

13. draft: compare candidates, source/fallback status, publishability, scorecard.

14. validation: inspect deterministic warnings, LLM findings, observations.

14. editorialCritiqueReport: inspect idea strength, tension, author stance, source integration, generic-prose risks, recommended editorial moves, candidate-level operationEnvelope, malformed JSON/schema incidents, and not-run incidents.

15. alternativeAngleTournament: inspect challenger route, candidate, model attempts, and why it entered or failed to enter final ranking.

16. rankingRevision: inspect pairwise winner, revision cycles, accepted/rejected moves.

Directed revision sub-results should include operationEnvelope for accepted, failed, or not-run writer revisions. revisionLoop.stopReason is canonical; inspect detailStopReason for legacy detail such as editorially-improved, no-fresh-angle, or provider failure detail. Inspect runtimeBudget to see max wall-clock, LLM calls, revision cycles, pairwise rounds, final repair cycles, non-improvement count, heartbeat, and slow-but-healthy status.

17. finalQualityGate: inspect the final quality contract, independent review attempts/model, final public-prose status, attributionReview, actionableAttributionFindings, diagnosticAttributionNoise, repair goals, accepted or rejected repair cycles, and final draft source.

18. qualityFidelity: inspect technicalStatus, providerRecoveryStatus, evidenceFidelity, issuelifecycle, editorialStatus, and overallVerdict before equating DraftRun.status=succeeded with publication-ready output. 18a. Cross-run reliability: when comparing multiple runs, use python scripts/analyze_draft_run_reliability.py --run-id <id> --run-id <id> and inspect operation/model counters, signalCoverage, plus the remediation ledger. Do not infer a systemic provider/model problem from one successful retry.

19. validation.progress: inspect nested operations; failed late operations should show safe errors and the final previous-best decision when available.

20. child AiRun detail: inspect prompt messages, role model, generation params, context pack, provider result, and sanitized raw response excerpt on JSON failures.

21. post-run editor state: inspect local postDraft.versions[] and finalText.editorDecisionSnapshot to see which version the human selected, which comments created new versions, and whether machine trace context was available when the final text was approved.

22. post-run quality check: for human-comment revisions, inspect postDraft.versions[].qualityCheck and child AiRun.requestPayload.draftRunStep = humanCommentRevisionQualityCheck to confirm matched/missed comment intents, source-marker preservation, public-prose status, internal jargon leaks, review attempts, incident metadata, and nested operationEnvelope on the revision/quality attempts.

23. post-run learning note: inspect authorNotes[] for type editorialLearning.

Pending and rejected notes should be visible but not used as author-position evidence. Accepted notes should flow through normal author-memory event and inference logic.

24. incident blast radius: when the same incidentType repeats across a run, compare it against CURRENT_LLM_OPERATION_INVENTORY to identify other migrated or allowlisted operations with the same expected incident coverage before deciding whether this is an isolated provider error or a systemic architecture issue.

25. payload budget blast radius: when qualityRisk, contextOverBudget, or payloadTooLarge repeats, compare payloadBudget.profileId, sent/trimmed counts, and inventory payloadBudgetStatus before blaming prompt quality.

10. Known AS IS limitations

- Editorial critique is active and feeds one alternative-angle challenger plus revision goals, but it is not yet an autonomous multi-agent debate loop.
- anotherAngle is active as one challenger route only, not as an autonomous debate loop or general fallback.
- ArticleDossier is a local compact memory artifact, not a vector store or cross-run RAG system.
- The deterministic context-access and typed dossier foundation is active for planning, writer/alternative-angle, review, ranking, revision, and final-quality runtime. Replay status readyForMigration proves factory/handle/exclusion behavior; runtime proof additionally requires child AiRun.requestPayload.providerDossier.runtimeMigrated=true. Slice 3.9 has final live acceptance in run 7bf3...; the separate alternativeAngleRoute over-budget finding was repaired and live-accepted in 3.9.1 by run 92532....
- Failed late provider operations are safely finalized when the Python call returns or raises. Evidence interpretation now has an operation-level timeout envelope; a broader infrastructure watchdog is still needed for externally stuck Celery tasks outside protected operation envelopes.
- The pipeline has detailed traceability. The main editor UI receives one machine draft first, then manages post-run human versions locally. Cross-post learning is currently limited to reviewable editorialLearning author-memory notes; promotion into actual editorial rules remains a future slice.

11. Maintenance rules

- This Markdown document and the PDF must be updated together.
- Regenerate the PDF after every edit:

```
python scripts/generate-draft-run-pipeline-pdf.py
```
- Future DraftRun planning, implementation, diagnostics, evaluation, and autofix work must read this document before changing or judging the pipeline.
- If current behavior differs from this document, update the document first or include the update in the same slice as the behavior change.